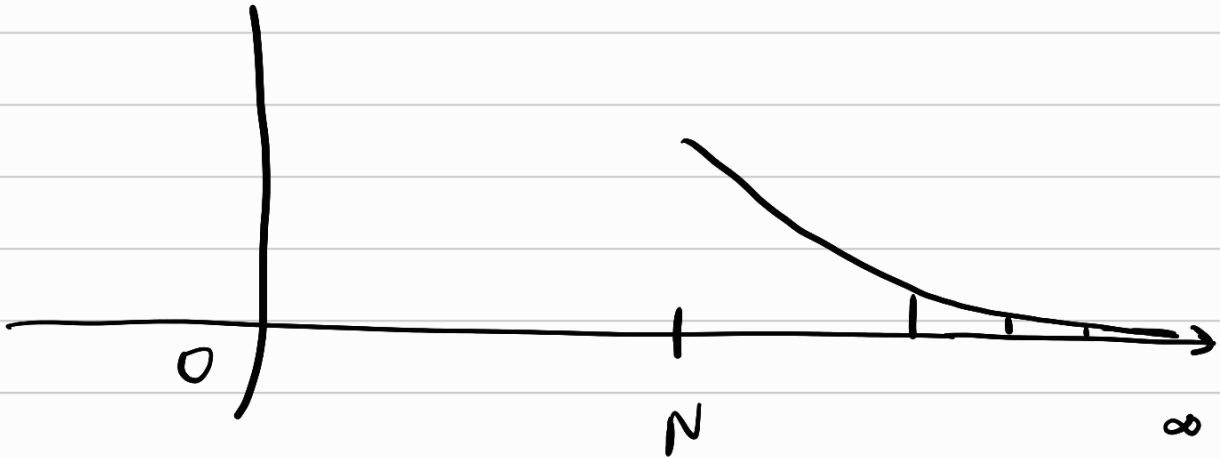


1. Show that $\lim_{N \rightarrow \infty} p(N) \int_N^\infty e^{-x^2} dx = 0 \quad \forall p \in \mathcal{P}$
 $p(x) \in \mathbb{R}[x]$.

Ans:



$$e^x = 1 + x + \frac{x^2}{2!} + \dots \quad \forall x \in \mathbb{R}$$

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \dots \quad \forall x \in \mathbb{R}$$

$$\Rightarrow e^{x^2} \geq \frac{(x^2)^{2n}}{n!} \quad \forall x \in \mathbb{R}, \forall n \in \mathbb{N}$$

$$\Rightarrow \frac{1}{e^{x^2}} \leq \frac{n!}{x^{4n}}$$

$$\Rightarrow x e^{-x^2} \leq \frac{(2n)!}{x^{4n}}$$

$$p(x) = p_0 + p_1 x + p_2 x^2 + \dots + p_n x^n$$

$$p(N) = p_0 + p_1 N + p_2 N^2 + \dots + p_n N^n$$

$$|p(N)| e^{-x^2} \leq \frac{|p(N)| 2^n!}{2^{2n}}$$

$$\leq \frac{(|p_0| + |p_1| N + \dots + |p_n| N^n)}{2^n!} \cdot \frac{1}{2^{2n}}$$

$$\int_N^\infty |p(N)| e^{-x^2} \leq \frac{2^n! (|p_0| + |p_1| N + \dots + |p_n| N^n)}{2^{2n}}$$

$$= \frac{2^n! (\dots)}{1 - 2n} x^{-2n+1} \Big|_N^\infty$$

$$= \frac{2^n! (\dots)}{1 - 2n} \left(-\frac{1}{N^{2n-1}} \right)$$

$$\leq \left(\frac{2^n!}{2^{2n-1}} \right) \frac{1}{N^{2n-1}} \quad N \geq n$$

$$\checkmark \quad 0 \leq \lim_{N \rightarrow \infty} \int_N^\infty \underbrace{|p(N)|}_{=0} \underbrace{e^{-x^2}}_{\text{und.}} \leq 0 \quad \checkmark \quad \textcircled{+}$$

$$\lim_{N \rightarrow \infty} \left| \int_N^\infty p(N) e^{-x^2} \right| \leq \lim_{N \rightarrow \infty} \underbrace{\int_N^\infty |p(N)| e^{-x^2}}_{=0}$$

$$\left| \lim_{N \rightarrow \infty} \int_N^\infty p(N) e^{-x^2} \right| = 0$$

$$\Rightarrow \lim_{N \rightarrow \infty} \int_N^\infty p(N) e^{-x^2} = 0.$$

Result: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is und. at $a \in \mathbb{R}$, and if $f(a) > 0$, then $\exists$ a nbh $(a-\delta, a+\delta)$ s.t. $\forall x \in (a-\delta, a+\delta)$

$$\underline{f(x) > 0}.$$

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \forall x \in (a-\delta, a+\delta)$$

$$\forall (x-a) < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$$

$$\boxed{f(a) - \varepsilon} < \underline{\underline{f(a)}} < f(a) + \varepsilon$$



$$f(x) = \begin{cases} x, & \forall x \in \mathbb{Q} \\ 1-x, & \forall x \in \mathbb{Q}^c \end{cases}$$